MATH 251: TOPOLOGY II EXAM 3

SPRING 2026

You have 80 minutes to do these problems in any order. There are 36 points total. Please show work unless otherwise specified.

You are allowed to look at Munkres, *Topology*, 2nd Ed., and any notes on paper that you wrote prior to the exam. You may omit proofs for statements proved in the textbook. You are not allowed to use electronic devices of any kind, including phones, computers, tablets, or audio devices. If you need to use the bathroom, please give any phones or other electronics to the proctor first. Throughout, $\mathbf{R}^2$ has the analytic topology and S^1 the subspace topology that it inherits from $\mathbf{R}^2$.

Problem 1 (6 points)**.** Recall that the set of integers $\mathbf{Z}$ forms a group under addition.

(a) Give a finite subset of $\mathbf{Z}$ that contains 2026 and generates $\mathbf{Z}$.
(b) Give an infinite subset of $\mathbf{Z}$ that contains 2026 and does not generate $\mathbf{Z}$.

You do not need to justify your answers.

Solution. (a) $\{1, 2026\}$. [Take any finite subset containing $a_1 = 2026$ and other integers $a_2, \ldots, a_k$ such that the greatest common divisor of the a_i is 1.]

(b) $2026\mathbf{Z}$. [Take any infinite subset containing 2026 whose elements are all divisible by an integer greater than 1.] $\square$

Problem 2 (6 points)**.** For each group presentation below, give a space whose fundamental group has that presentation. You do not need to justify your answers.

(a) $\langle a, b \rangle$.
(b) $\langle a, b \mid aba^{-1}b^{-1} \rangle$.
(c) $\langle a \mid a^2 \rangle$.

Solution. (a) $S^1 \vee S^1$. [The fundamental group is freely generated by the path-homotopy classes of loops homotopic to the two circles: See Munkres §71.]

(b) $S^1 \times S^1$. [The fundamental group is isomorphic to $\mathbf{Z} \times \mathbf{Z}$, which has the presentation in (b): See Munkres §72.]

(c) The real projective plane P^2. [The fundamental group is $\mathbf{Z}/2\mathbf{Z}$, which has the presentation in (c): See Munkres §60 and §72.] $\square$

Problem 3 (7 points)**.** For each space S below, decide whether the complement of a simple closed curve $C \subseteq S$ is always disconnected. If not, state an example where $S - C$ is connected. You do not need to justify the example.

(a) S^2.
(b) $S^1 \times S^1$.
(c) $\mathbf{R}^2$.

Solution. (a) Yes, by Jordan separation.

(b) No. Fix a point $p \in S^1$ and take $C = S^1 \times \{p\}$.

(c) Yes, by Jordan separation. □

Problem 4 (7 points)**.** Let $A \subseteq X$ be path connected. Recall that a retraction of X onto A is a continuous map $r\colon X \to A$ such that $r(a) = a$ for all $a \in A$.

(a) Show that if $r\colon X \to A$ is a retraction and X is simply connected, then A is simply connected.

(b) Is there a retraction from $\mathbf{R}^2$ onto the unit circle S^1? Justify your answer.

Solution. (a) Let $i\colon A \to X$ be the inclusion. Then $r \circ i$ is the identity map of A, so for any basepoint $a \in A$, the homomorphism

$$r_* \circ i_* = (r \circ i)_*\colon \pi_1(A, a) \to \pi_1(A, a)$$

must be the identity homomorphism. This is bijective, so r_* must be surjective. So if the fundamental group of X is trivial, then the fundamental group of A is trivial. As A is also path connected, it must be simply connected.

(b) No. Indeed, $\mathbf{R}^2$ is simply connected, being contractible, whereas S^1 is path connected but not simply connected. So the existence of a retraction from $\mathbf{R}^2$ onto S^1 would contradict (a). □

Problem 5 (6 points)**.** Let $S = \{x, y\}$, and let

$$\varphi\colon F_S \to \mathbf{Z} \times \mathbf{Z}$$

be a homomorphism.

(a) Show that $\varphi(xyx^{-1}y^{-1}) = (0, 0)$.

(b) Suppose that $\varphi(x) = (1, 0)$ and $\varphi(y) = (0, 1)$. Is φ surjective? (Yes/No)

(c) Suppose that $\varphi(x) = (1, 0)$ and $\varphi(y) = (1, 1)$. Is φ surjective? (Yes/No)

Solution. (a) We compute

$$\begin{aligned}
\varphi(xyx^{-1}y^{-1}) &= \varphi(x) + \varphi(y) + \varphi(x^{-1}) + \varphi(y^{-1}) && \text{since } \varphi \text{ is a homomorphism}\\
&= \varphi(x) + \varphi(y) - \varphi(x) - \varphi(y) && \text{since } \varphi \text{ is a homomorphism}\\
&= \varphi(x) - \varphi(x) + \varphi(y) - \varphi(y) && \text{since addition is commutative}\\
&= (0, 0).
\end{aligned}$$

(b) Yes. [For any $a, b \in \mathbf{Z}$, we have $\varphi(x^a y^b) = (a, b)$.]

(c) Yes. [Observe that $\varphi(yx^{-1}) = \varphi(y) - \varphi(x) = (0, 1)$. So we reduce to the situation in (b).] □

Problem 6 (4 points)**.** Fix a basepoint $p \in S^1 \vee S^1$. Show that for any continuous map $f\colon S^1 \vee S^1 \to S^1 \times S^1$, the kernel of the homomorphism

$$f_*\colon \pi_1(S^1 \vee S^1, p) \to \pi_1(S^1 \times S^1, f(p))$$

cannot be trivial. You may assume Problem 5(a).

Solution. Recall that $\pi_1(S^1 \vee S^1, p) \simeq F_2$ for any $p \in S^1 \vee S^1$, whereas $\pi_1(S^1 \times S^1) \simeq \mathbf{Z} \times \mathbf{Z}$ for any $q \in S^1 \times S^1$. Therefore, by Problem 5(a), the kernel contains a nonidentity element. $\square$